Agentic Graph Reasoning

Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models

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Pre-defense

The problem

Language models answer factual questions fluently.

They answer just as fluently when they do not hold the fact.

Nothing in the wording separates the two.

The gap

Fluency and factuality are produced by the same mechanism, so the output gives the reader no signal about which one they are getting.

A system that is confidently wrong is worse than one that says it does not know.

The failure is not visible in the answer.

How often? Our own control

No retrieval, same backbone, WebQSP:

- 661 entities asserted
- 179 of them do not exist in the knowledge graph
- that is 27.1%, stated with no hedge

Read that again

More than one asserted entity in four names nothing at all — and the sentence around it is indistinguishable from the ones that are right.

The failure is not rare either: 27.1%, measured here rather than quoted.

What a hallucination is

Output that is fluent, well formed, and backed by no source the model can point to.

Factually wrong

It contradicts the world. *"Ada Lovelace was born in Paris."* Checkable — if you already have a reference to check it against.

Unsupported

It may even be true, but nothing the system retrieved says so. Checkable without an oracle: against what was retrieved.

From the outside the two are identical: confident, well formed, no hedge.

A retrieval system can only police the second — so that is the one this thesis attacks, and the only one it claims.

Where it comes from

Not a bug that escaped testing — a consequence of how these models are built, which is why the response has to be architectural.

The training data

Errors and thin coverage, reproduced faithfully.

The model

Knowledge is spread across the weights with no index, so nothing separates a memorised fact from a plausible continuation.

The prompt

What we put in the context window — and what we leave out of it.

Generating text and recalling a fact are the same operation.

Which of the three can we act on?

Only the third

The other two need a curated corpus or a retrained model, and neither is available to most people deploying one of these systems.

Controlling what enters the context window — and checking what comes back out against it — is a systems problem. That is this thesis.

And multi-hop compounds it

When an answer needs two facts chained, a wrong first step is never revisited: the rest of the chain proceeds from a false premise, just as fluently.

Fix what goes in, then check what comes out against it — before anything is emitted.

What a knowledge graph is

Facts are stored as triples — head, relation, tail:

(Ada Lovelace, place_of_birth, London)

A question needing two of them *chained* is multi-hop.

Why a graph at all

Every fact has an address. A claim can then be checked by looking for an edge — rather than by asking the model whether it believes itself.

Each step depends on the last, so a wrong first hop is never seen again.

The complication: mediator nodes

Freebase stores any fact with more than two participants as a node, not an edge.

"Rainn Wilson played Dwight in The Office" is one node joining three things — series, character, actor.

What that costs

63.7% of this graph's nodes are mediators, and they carry no name.

A path that reads as two hops is often three.

Nothing in the graph is called plays: the relation a question names is rarely the relation the graph stores.

Where existing approaches stop

Approach	What it does	Where it stops
Parametric	Answers from weights	No external evidence at all
Vector RAG	Retrieves text once, then reasons	Retrieval happens *before* reasoning begins
Static GraphRAG	Retrieves a fixed neighbourhood	Radius fixed before the question is understood
Agentic navigation	Interleaves retrieval and reasoning	Emits whatever the last generation call produces

Nobody checks what the answer asserts *before* it is delivered.

What everyone else does

System	Exploration	Decomp.	Backtrack	Verifies first
GraphRAG	Fixed radius, no search	No	No	No
RoG	Generated relation paths	Partial	No	No
ToG	LLM-guided beam search	No	No	No
PoG	Adaptive planning	Yes	Yes	Partial
AGR	**Guided beam, backtracking**	**Yes**	**Yes**	**Yes**

Their published Hits@1 runs higher than anything here and is *not* comparable: other backbones, other subsets, full Freebase.

No prior system checks its *answer's* claims against what it retrieved, before answering.

Research questions

RQ1 Does agentic navigation improve multi-hop factual accuracy — and does the advantage grow with hop count?

RQ2 Given a system that already navigates the graph, what does a claim-level check against traversed triples contribute?

RQ3 Which components earn their cost, in accuracy and in tokens?

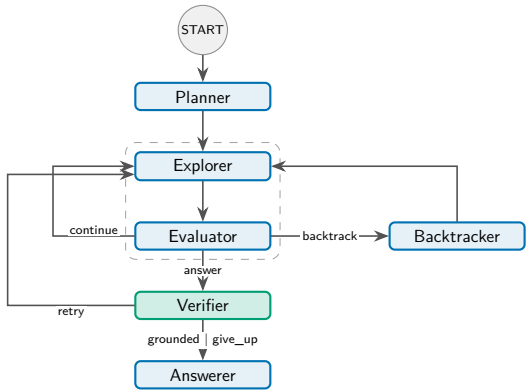
RQ2 is posed as *what does it contribute*, not *does it reduce hallucination*. The answer has several parts, and a yes/no framing would hide them.

AGR: an explicit state machine

Six nodes; the model is consulted only at decision points.

- **Planner** — sub-objectives
- **Explorer** — expands the frontier
- **Evaluator** — objective met?
- **Backtracker** — undo + ban list
- **Verifier** — the contribution
- **Answerer** — emits what survived

Three cycles — *continue*, *backtrack*, *retry* — all bounded by budgets rather than by model behaviour.



Constrained tools, not free-form queries

Operation	Returns	Why it is bounded
`get_relations`	Candidate relations	≤ 300 offered to the scorer
`get_neighbors`	Neighbours of an entity	≤ 200 per expansion
`verify_connection`	Adjacency in the full graph	Used only by the verifier
`search_entity`	Surface form $\rightarrow$ node	Three-stage resolver

The agent never writes Cypher; every operation is logged.

Determinism in the tools is what makes the verification layer possible.

The Structural Verification Layer

Before anything is emitted:

1. Draft answer is split into atomic claims
2. Each claim is checked against the triples actually traversed
3. Claims grounded by the traversal keep those triples as evidence
4. Unsupported claims trigger targeted re-exploration, or are dropped

The behaviour this buys

The system **hedges rather than asserts**, and carries its evidence with the answer.

This is the contribution. The next slide is what it does *not* do.

What *structural* means — and what it does not

What it means

The check is against the graph the agent walked, not the model's judgement of its own output.

What it does not mean

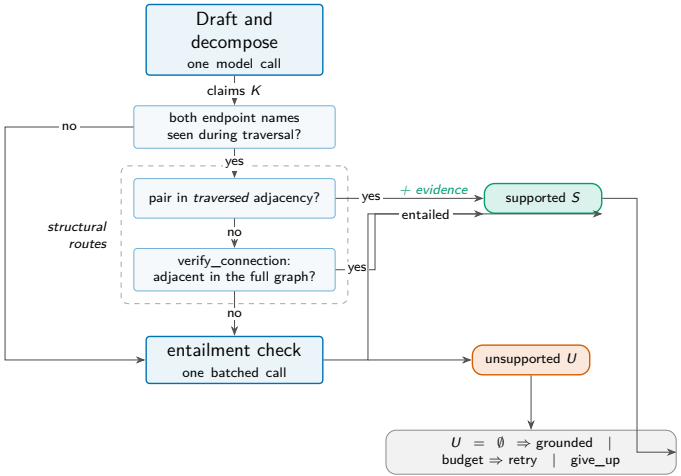
It tests adjacency: whether an edge joins the claim's endpoints, not whether it is the edge the claim *names*, and not its direction. A *mother* claim passes on a *child* edge — only route three reads the relation.

The other honest limit

A claim can be *true* and still be the wrong answer. Structural grounding cannot catch that — see the echo attractor later.

"Its evidence" is narrower than it sounds, too: one route of three records any, and only inside the run.

One claim, three routes



Only the third route costs a model call. The first two are pure graph lookups — and neither of them reads the relation.

One question, end to end

WebQTest-960 — "who plays dwight in the office"

1. Planner — two sub-objectives; The Office resolved to one node
2. Explorer — scores relations, takes tv.tv_program.regular_cast
3. Evaluator — objective met, resolves Dwight Schrute
4. Explorer — takes tv.regular_tv_appearance.character
5. Evaluator — resolves Rainn Wilson, decides answer
6. Verifier — 2 claims, 2 supported, none dropped

What came out

"Rainn Wilson plays Dwight Schrute in The Office."

6 model calls · depth 2 · 18 supporting triples

The second hop runs *through* a mediator — which is why that relation is named for the *appearance*, not for plays.

The environment and the question sets

Freebase-derived graph

Property	Value
Entities	2,592,892
Triples	8,309,194
Distinct relations	7,058
Import time	36.4 s

Certified question sets

	WebQSP	CWQ
Questions	400	400
Gold (median)	1.5	1.0
Reachable	97.0%	99.2%
Multi-hop	132	260

Reachability is the *ceiling* on every Hits@1 reported: a miss is the system's, not the environment's.

800 **questions, with reachability certified per question so the accuracy ceiling is known.**

Making the comparison fair

Five systems, one frozen backbone, one budget:

- No-retrieval (parametric control)
- Vector RAG
- Static GraphRAG
- Think-on-Graph (agentic)
- **AGR**

Held constant

Same model, temperature 0, same 25-call cap, same questions, same graph.

What this buys

Differences are attributable to **architecture**, not to model capacity or spend.

One thing is *not* held equal, and it is the next slide.

What is *not* held equal

Candidate widths

ToG prunes to 40/20 candidates per step, AGR to 300/200.

Narrower is cheaper, so it cannot explain ToG's clipping — but its unclipped score is a lower bound.

Why a parametric control

These benchmarks predate current models and may be partly memorised. Without it we could not separate retrieval from recall.

Said here rather than found later: this is limitation five in the thesis.

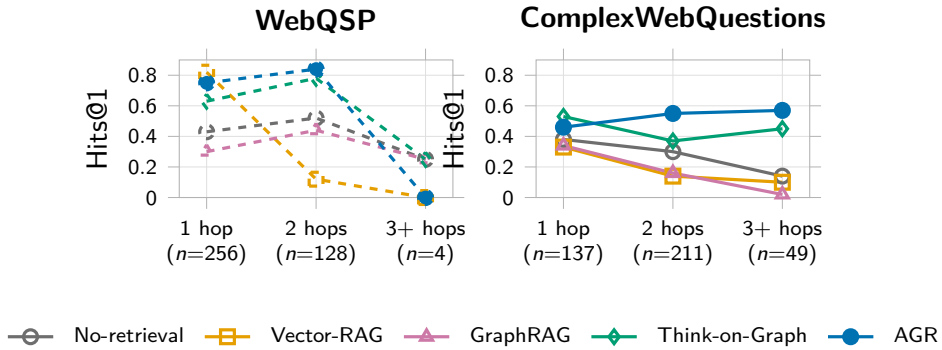
Main results

	WebQSP		ComplexWebQuestions		Cost / question	
System	**Hits@1**	**F1**	**Hits@1**	**F1**	**Tokens**	**Calls**
No-retrieval	0.453	0.305	0.307	0.279	113	1.0
Vector RAG	0.568	0.432	0.203	0.168	527	1.0
Static GraphRAG	0.340	0.262	0.205	0.183	531	1.0
Think-on-Graph	0.660	0.477	0.435	0.378	3,615	12.8
AGR	**0.755**	**0.642**	**0.522**	**0.469**	4,511	**6.2**

Cost is WebQSP. CWQ: AGR 6,818 tokens, 8.9 calls; Think-on-Graph 5,572, 18.2.

Ahead of every baseline on both datasets, at half the model calls of the other agentic system.

RQ1: accuracy against hop count



On CWQ, AGR goes 0.46 → 0.55 → 0.57 — the **only** system that ends above where it started. Three of the other four decay monotonically.

RQ1 is answered by a shape, not a difference of means.

The caveat I want to raise myself

	WebQSP		ComplexWebQuestions	
Hits@1 on...	**ToG**	**AGR**	**ToG**	**AGR**
questions ToG *finishes*	**0.852**	0.788	**0.629**	0.607
questions ToG is *clipped* on	0.197	**0.675**	0.188	**0.415**
clip rate	29.2% (117/400)		44.0% (176/400)	

Where Think-on-Graph finishes inside the shared 25-call cap, it is **ahead of AGR on both datasets**. The entire aggregate margin comes from the questions it cannot finish.

The claim is efficiency under a fixed budget — not that AGR reasons better per step.

RQ2: does anything ungrounded get asserted?

System	Entities asserted	Ungrounded	Rate
No-retrieval	1,001	221	22.1%
Vector RAG	1,329	45	3.4%
Static GraphRAG	1,024	8	0.8%
Think-on-Graph	1,265	0	**0.0%**
AGR	1,709	0	**0.0%**

Both datasets pooled. AGR asserts no entity that does not exist in the graph, across 1,709 assertions.

But Think-on-Graph reaches 0.0% too — so this is a property of *graph navigation*, not of the verification layer.

RQ2: what verification does *not* do

It does not move accuracy

Removing it changes accuracy by an amount we cannot detect: $p = 1.0$ on *both* datasets. It does not earn its place on Hits@1 or F1.

The record does not survive the run

The log keeps the count of supporting triples and discards the list. *Auditable in the run, not from the record.*

This is a negative result reported as one. The layer's case rests on auditability, and the thesis bounds that too rather than claiming an accuracy benefit the measurement does not support.

RQ2: so what does it do?

It withholds what it cannot ground

Removing the layer drops hedging 23.2% → 20.2% on CWQ and 8.5% → 8.0% on WebQSP. A hedge is a calibrated non-assertion, not a refusal.

It attaches supporting triples

From one route of three: verify_connection and entailment attach none.

It pairs answer with evidence

In the run — which is where the previous slide's bound bites.

Silent error becomes an explicit hedge. That is the claim, and it is the whole of it.

RQ3: what each component earns

	WebQSP		ComplexWebQuestions		
Condition	**F1**	p	**F1**	p	**Tokens (WQ)**
Full system	0.659	—	0.445	—	4,562
No planner	**0.742**	**0.006**	0.398	0.088	**3,159**
No backtracking	0.646	0.727	0.445	1.000	4,261
No verifier	0.663	1.000	0.436	1.000	4,460
Embedding-only scoring	0.643	0.664	0.437	0.481	3,413

Paired McNemar, halves of 200 and 198.

Three of the four change nothing detectable. The fourth is on the next slide, and its sign is backwards.

RQ3: one effect, and its sign is backwards

Removing the planner *improves* WebQSP

$+0.083$ F1, $p = 0.006$, and -31% tokens.

ComplexWebQuestions trends the other way: -0.047, $p = 0.088$.

What follows

WebQSP is mostly one hop, and decomposing a one-hop question sends the frontier away from the answer.

Gate decomposition on structure — a conclusion the measurement forced, not one it confirmed.

Every failure, read and labelled

All 259 remaining failures were read and labelled by hand, both datasets pooled:

Category	n
Relation selection	65
Composite claim	47
Knowledge-graph gap	44
Decomposition error	38
Answer selection	15
Echo attractor	13

These are totals, and totals hide things

Wrong and hedge are *never pooled* in the thesis, and the shape flips across datasets: composite claim is 1 on WebQSP against 46 on CWQ.

Split census: backup page 4.

The largest category is relation selection — the agent taking the wrong edge, not inventing one.

The characteristic error of a graph navigator is *not* invention.

The echo attractor

Sixth in that census, with 13 cases, and the one worth naming:

a real, grounded entity one hop from the answer.

Ask for a director and get the film. Ask for a capital and get the country.

Why it matters

Any grounding check *passes* it — because it is true. The entity is real, it was traversed, the triple exists. Verification cannot catch it by construction.

Different systems fall into it together, so no evaluation treating them as independent can see it — and a policy of rescoring on majority agreement turns it into apparent correctness.

The contribution is the mechanism, named, not the count.

Contributions, and what I would not claim

Contributions

1. AGR and its Structural Verification Layer
2. A component-level ablation of four mechanisms
3. Stratum-dependent decomposition — planning hurts
4. The *echo attractor*, named
5. Benchmark-defect rates for WebQSP and CWQ (57)
6. Pre-specified evaluation thresholds

Limitations I state plainly

- The verifier logs only what it rejects: wrongful acceptance is unmeasured, and the evidence is not persisted
- No detectable accuracy gain — and at $n \approx 200$ that is "not detected", not "none"
- Zero ungrounded assertion is navigation, not the layer
- One environment, one backbone, one annotator
- ToG leads where it finishes, from a narrower candidate set: 40/20 vs 300/200

Thank you

Questions